LIDARLearn: A Unified Deep Learning Library for 3D Point Cloud Classification, Segmentation, and Self-Supervised Representation Learning

Said Ohamouddou said_ohamouddou@um5.ac.ma
ENSIAS, Mohammed V University, Rabat, Morocco

Hanaa El Afia
ENSIAS, Mohammed V University, Rabat, Morocco

Abdellatif El Afia
ENSIAS, Mohammed V University, Rabat, Morocco

Raddouane Chiheb
ENSIAS, Mohammed V University, Rabat, Morocco

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Abstract

Three-dimensional (3D) point cloud analysis has become central to applications ranging from autonomous driving and robotics to forestry and ecological monitoring. Although numerous deep learning methods have been proposed for point cloud understanding, including supervised backbones, self-supervised pre-training (SSL), and parameter-efficient fine-tuning (PEFT), their implementations are scattered across incompatible codebases with differing data pipelines, evaluation protocols, and configuration formats, making fair comparisons difficult. We introduce LIDARLearn, a unified, extensible PyTorch library that integrates over 55 model configurations covering 29 supervised architectures, seven SSL pre-training methods, and five PEFT strategies, all within a single registry-based framework supporting classification, semantic segmentation, part segmentation, and few-shot learning. LIDARLearn provides standardised training runners, cross-validation with stratified K-fold splitting, automated LaTeX/CSV table generation, built-in Friedman/Nemenyi statistical testing with critical-difference diagrams for rigorous multi-model comparison, and a comprehensive test suite with 2 200+ automated tests validating every configuration end-to-end. The code is available at `https://github.com/said-ohamouddou/LIDARLearn` under the MIT licence.

Keywords: Point Clouds, 3D Deep Learning, Self-Supervised Learning, Parameter-Efficient Fine-Tuning, Open Source Software

1 Introduction

Light Detection and Ranging (LiDAR) technology has become a cornerstone of 3D scene understanding across scientific and engineering disciplines. In forestry, airborne and terrestrial laser scanning generate complex point cloud datasets that require advanced processing to extract structural metrics, such as canopy height and above-ground biomass **?**. In

autonomous driving and indoor robotics, real-time semantic segmentation of dense point clouds is critical for navigation and scene understanding Wu et al. (2024). The emergence of deep learning methods that operate directly on unstructured point sets, beginning with PointNet Qi et al. (2017a) and PointNet++ Qi et al. (2017b), has driven rapid progress, spawning dozens of architectures that exploit local geometry Wang et al. (2019); Xiang et al. (2021), attention mechanisms Guo et al. (2021); Zhao et al. (2021); Wu et al. (2022, 2024), self-supervised pre-training Pang et al. (2022); Yu et al. (2022); Chen et al. (2023); Dong et al. (2023); Qi et al. (2023), and parameter-efficient transfer Zha et al. (2023); Zhou et al. (2024); Sun et al. (2024); Liang et al. (2024).

Despite this progress, the practical impact is limited by fragmentation; each method ships in its own repository with bespoke data loaders, evaluation scripts, and configuration formats. Reproducing or extending published results requires significant engineering efforts to reconcile these differences. Existing open-source libraries partially address this—OpenPoints **?** supports ∼15 classification backbones, Pointcept **?** focuses on segmentation, and Torch-Points3D **?** provides a modular framework—but none unifies supervised training, SSL pre-training, and PEFT fine-tuning across all three task types (classification, semantic segmentation, and part segmentation) under a single consistent interface.

Here, we introduce LIDARLearn, a comprehensive library that fills this gap. LIDARLearn offers: (i) 29 supervised classification/segmentation backbones and 7 SSL pre-training methods faithfully ported from their original repositories and verified by dual code audits; (ii) 5 PEFT strategies applicable to 4 SSL backbones, yielding 20 additional fine-tuning configurations; (iii) standardised training pipelines supporting classification, semantic segmentation, part segmentation, and few-shot learning; (iv) rigorous evaluation infrastructure including stratified K-fold cross-validation, per-class metrics via `scikit-learn`, and automated LaTeX table generation; and (v) a 2 200-test automated test suite covering configuration parsing, forward-pass shape verification, pre-trained checkpoint loading, and script coverage. We demonstrate the library by benchmarking all 56 configurations on a LiDAR tree species classification task (Appendix C).

2 Advantages and Comparison to Existing Toolkits

Table 1 compares LIDARLearn with five established point cloud libraries across key capability dimensions.

Model Diversity. LIDARLearn integrates 29 supervised backbones, 7 SSL pre-training methods, and 5 PEFT strategies. The supervised backbone spans three families. *Point-based methods:* DELA Chen et al. (2024), PPFNet Deng et al. (2018), PointCNN Li et al. (2018b), PointConv Wu et al. (2019), PointKAN Stanford (2024), PointMLP Ma et al. (2022), PointNet Qi et al. (2017a), PointNet++ Qi et al. (2017b), PointSCNet Chen et al. (2022), PointWeb Zhao et al. (2019), RSCNN Liu et al. (2019), RepSurf Ran et al. (2022), and SONet Li et al. (2018a). *Attention and transformer architectures:* GlobalTransformer Berg et al. (2022), P2P Wang et al. (2022), PCT Guo et al. (2021), PVT Zhang et al. (2021), PointTNT Berg et al. (2022), PointTransformer Zhao et al. (2021), PointTransformerV2 Wu et al. (2022), and PointTransformerV3 Wu et al. (2024). *Graph neural networks:* CurveNet Xiang et al. (2021), DGCNN Wang et al. (2019), DeepGCN Li et al. (2019), GDAN Xu et al. (2021), KANDGCNN Ohamouddou et al. (2026b), MS-

Table 1: Comparison of LIDARLearn with existing point cloud deep learning libraries. *Cls* = classification models, *Seg* = semantic segmentation models, *Part* = part segmentation models.

Feature	**Learning3D**	**Open3D-ML**	**OpenPoints**	**Pointcept**	**Torch-Points3D**	**LIDARLearn**
Cls models	5	0	15	3	6	**29**
Seg models	2	5	9	11	10	**26**
Part-seg models	0	0	5	1	2	**26**
SSL pre-training	0	0	1	1	0	**7**
PEFT strategies	0	0	0	0	0	**5**
Few-shot eval	—	—	—	—	—	✓
K-fold CV	—	—	—	—	—	✓
Automated tests	—	—	—	—	—	✓

DGCNN Zhai et al. (2021), and MSDGCNN2 Ohamouddou et al. (2026a). For *self-supervised pre-training*, seven methods are included: ACT Dong et al. (2023), PCP-MAE Zhang et al. (2024), Point-BERT Yu et al. (2022), PointGPT Chen et al. (2023), Point-M2AE Zhang et al. (2022), Point-MAE Pang et al. (2022), and ReCon Qi et al. (2023). Five *parameter-efficient fine-tuning* strategies are supported: DAPT Zhou et al. (2024), IDPT Zha et al. (2023), PPT Sun et al. (2024), PointGST Liang et al. (2024), and VPT Jia et al. (2022) were integrated with four SSL backbones (PointMAE, PointGPT, ACT, and ReCon), producing 20 additional fine-tuning configurations.

SSL Faithfulness. Each SSL implementation was audited against its original repository in two independent rounds, verifying the core model logic, loss computation, checkpoint prefix-stripping, freezing behavior, and classification/segmentation output consistency. All critical issues found during the audits were fixed and re-verified (see Appendix B).

Evaluation Rigour. LIDARLearn implements stratified K-fold cross-validation with per-fold model rebuild, optimizer reset, and deterministic seeding—preventing state leakage between folds. Metrics (accuracy, balanced accuracy, macro/weighted F1, Cohen's κ, per-class precision/recall) were computed via `scikit-learn` at the best-validation-accuracy epoch. Few-shot evaluation follows the standard 10-episode protocol on ModelNet40 with automated mean±std aggregation. The results were exported as LaTeX tables with per-column best-value highlighting and CSV files for programmatic analysis.

3 Toolbox Usage

3.1 Dependencies and Installation

LIDARLearn requires Python ≥3.8, PyTorch ≥1.10, and `timm`. The required CUDA extensions (`chamfer_dist`, `pointnet2_ops`, `dela_cutils`) are built via `scripts/install_extensions.sh`. The library uses a YAML-based configuration system with `_base_` inheritance, where the dataset parameters, model hyperparameters, and optimizer settings are cleanly separated.

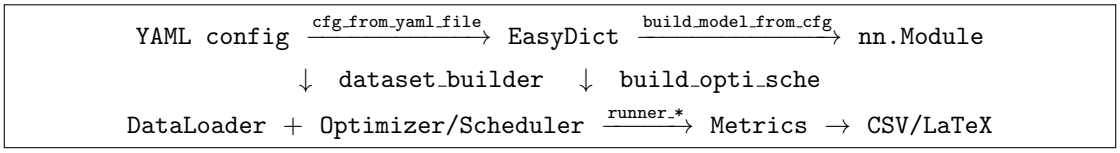

Figure 1: Simplified LIDARLearn pipeline. YAML configs drive model construction, dataset loading, and optimizer building via a shared registry. Task-specific runners produce standardised metric outputs.

3.2 Workflow

The following example trains PointMAE with DAPT fine-tuning on a custom LiDAR tree species dataset using 5-fold cross-validation.

```
# 1. Pre-train on ShapeNet55 (300 epochs)
python main.py --config cfgs/classification/PointMAE/ShapeNet55/
    shapenet55_pretrain.yaml \
--mode pretrain --exp_name mae_pretrain

# 2. Fine-tune with DAPT on STPCTLS (5-fold CV)
python main.py --config cfgs/classification/PointMAE/STPCTLS/
    stpctls_cv_dapt.yaml \
--ckpts pretrained/pretrained_mae.pth --run_all_folds --seed 42

# 3. Generate LaTeX comparison table
python scripts/generate_comparison_table_one_table.py \
--exp_dir experiments/STPCTLS
```

All 56 classification model/strategy configurations can be launched using a single command: `bash scripts/train_stpctls.sh`. Equivalent sweep scripts exist for ModelNet40, S3DIS segmentation, ShapeNetParts part segmentation, and ModelNet40 few-shot evaluation. For rigorous model comparison across k-fold cross-validation runs, LIDARLearn also provides `scripts/friedman_cv_analysis.py`, which performs the Friedman non-parametric test with Nemenyi post-hoc analysis and critical-difference diagrams over the per-fold scores of all models, producing a publication-ready LaTeX output.

3.3 Architecture

Figure 1 illustrates this pipeline. Models are registered via `@MODELS.register_module()` decorators and built from YAML configs through a unified `build_model_from_cfg` factory. Five task-specific runners (`runner_finetune`, `runner_pretrain`, `runner_seg`, etc.) share a common infrastructure for checkpointing, metrics tracking, and logging, whereas each implements task-specific forward/loss logic. The PEFT optimizer builder (`_build_peft_optimizer`) automatically freezes the backbone parameters and trains only strategy-specific adapters/prompts based on a single `optimizer.part` config field.

3.4 Adding a New Model

Integrating a new classification backbone requires three steps.

1. **Model file.** Create `models/mymodel/mymodel.py` with a class that inherits from `BasePointCloudModel` and is decorated with `@MODELS.register_module()`. The class must implement `forward(pts)` returning `[B, num_classes]` logits; `get_loss_acc` is inherited automatically.
2. **Registration.** Add `import models.mymodel.mymodel` to `models/__init__.py` so the registry discovers the class at import time.
3. **Config.** A YAML file (for example, `cfgs/classification/MyModel/STPCTLS/stpctls.yaml`) specifying `model.NAME: MyModel` alongside dataset, optimizer, and scheduler fields. The model is now trainable via `python main.py --config <yaml> --mode finetune`.

Segmentation models follow the same pattern but inherit from `BaseSegModel` and implement `forward(pts, cls_label)` returning the per-point logits. No changes to the runners, metrics, or testing infrastructure are needed—the new model is automatically covered by the L1 config parse test suite.

4 Conclusion

We presented LIDARLearn, an open-source library that unifies 29 supervised backbones, seven SSL pre-training methods, and five PEFT strategies for 3D point cloud analysis under a single, tested, and extensible framework. LIDARLearn supports classification, semantic segmentation, part segmentation, and few-shot evaluation with rigorous cross-validation and automated reporting. With 2 200+ automated tests and dual-round SSL implementation audits, the library is designed for reproducible research and fair benchmarking. As a case study, we benchmarked all 56 model configurations on the STPCTLS terrestrial laser scanning dataset for LiDAR tree species classification (Appendix C), demonstrating the library's ability to conduct comprehensive and standardized comparisons across supervised, SSL, and PEFT methods for a real-world domain-specific task.

References

Axel Berg, Magnus Oskarsson, and Mark O'Connor. Points to patches: Enabling the use of self-attention for 3d shape recognition. In *International Conference on Pattern Recognition (ICPR)*, 2022.

Binjie Chen, Yunzhou Xia, Yu Zang, Cheng Wang, and Jonathan Li. Decoupled local aggregation for point cloud learning. *arXiv preprint arXiv:2308.16532*, 2024.

Guangyan Chen, Meiling Wang, Yi Yang, Kai Yu, Li Yuan, and Yufeng Yue. Pointgpt: Auto-regressively generative pre-training from point clouds. In *Advances in Neural Information Processing Systems*, volume 36, pages 29615–29627, 2023.

Xingye Chen, Yiqi Wu, Wenjie Xu, Jin Li, Huaiyi Dong, and Yilin Chen. Pointscnet: Point cloud structure and correlation learning based on space-filling curve-guided sampling. *Symmetry*, 14(1):8, 2022.

Haowen Deng, Tolga Birdal, and Slobodan Ilic. Ppfnet: Global context aware local features for robust 3d point matching. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

Runpei Dong, Zekun Qi, Linfeng Zhang, Junbo Zhang, Jianjian Sun, Zheng Ge, Li Yi, and Kaisheng Ma. Autoencoders as cross-modal teachers: Can pretrained 2d image transformers help 3d representation learning? In *International Conference on Learning Representations*, 2023.

Meng-Hao Guo, Jun-Xiong Cai, Zheng-Ning Liu, Tai-Jiang Mu, Ralph R Martin, and Shi-Min Hu. Pct: Point cloud transformer. *Computational Visual Media*, 7:187–199, 2021.

Menglin Jia, Luming Tang, Bor-Chun Chen, Claire Cardie, Serge Belongie, Bharath Hariharan, and Ser-Nam Lim. Visual prompt tuning. In *European Conference on Computer Vision*, pages 709–727, 2022.

Guohao Li, Matthias Müller, Ali Thabet, and Bernard Ghanem. Deepgcns: Can gcns go as deep as cnns? In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.

Jiaxin Li, Ben M. Chen, and Gim Hee Lee. So-net: Self-organizing net for point cloud analysis. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018a.

Yangyan Li, Rui Bu, Mingchao Sun, Wei Wu, Xinhan Di, and Baoquan Chen. Pointcnn: Convolution on x-transformed points. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018b.

Dingkang Liang, Tianrui Feng, Xin Zhou, Yumeng Zhang, Zhikang Zou, and Xiang Bai. Parameter-efficient fine-tuning in spectral domain for point cloud learning. *arXiv preprint arXiv:2410.08114*, 2024.

Yongcheng Liu, Bin Fan, Shiming Xiang, and Chunhong Pan. Relation-shape convolutional neural network for point cloud analysis. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

Xu Ma, Can Qin, Haoxuan You, Haoxi Ran, and Yun Fu. Rethinking network design and local geometry in point cloud: A simple residual mlp framework. In *International Conference on Learning Representations (ICLR)*, 2022.

Said Ohamouddou, Hanaa El Afia, Mohamed Hamza Boulaich, Abdellatif El Afia, and Raddouane Chiheb. Ms-dgcnn++: Multi-scale dynamic graph convolution with scale-dependent normalization for robust lidar tree species classification, 2026a. URL `https://arxiv.org/abs/2507.12602`.

Said Ohamouddou, Mohamed Ohamouddou, Hanaa El Afia, Abdellatif El Afia, Rafik Lasri, and Raddouane Chiheb. Introducing the short-time fourier kolmogorov arnold network: A dynamic graph cnn approach for tree species classification in 3d point clouds. *Pattern Recognition*, 165:112584, 2026b.

Yatian Pang, Wenxiao Wang, Francis EH Tay, Wei Liu, Yonghong Tian, and Li Yuan. Masked autoencoders for point cloud self-supervised learning. In *European Conference on Computer Vision*, pages 604–621, 2022.

Charles R Qi, Hao Su, Kaichun Mo, and Leonidas J Guibas. Pointnet: Deep learning on point sets for 3d classification and segmentation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 652–660, 2017a.

Charles Ruizhongtai Qi, Li Yi, Hao Su, and Leonidas J Guibas. Pointnet++: Deep hierarchical feature learning on point sets in a metric space. In *Advances in Neural Information Processing Systems*, volume 30, 2017b.

Zekun Qi, Runpei Dong, Guofan Fan, Zheng Ge, Xiangyu Zhang, Kaisheng Ma, and Li Yi. Contrast with reconstruct: Contrastive 3d representation learning guided by generative pretraining. In *International Conference on Machine Learning*, pages 28223–28243, 2023.

Haoxi Ran, Jun Liu, and Chengjie Wang. Surface representation for point clouds. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

Dominik Seidel, Peter Annighöfer, Martin Ehbrecht, Delphine C Zemp, Setotaw Alem, and Tobias Hedrich. Predicting tree species from 3d laser scanning point clouds using deep learning. *Frontiers in Plant Science*, 12:635440, 2021.

Ali Stanford. Pointnet with kan versus pointnet with mlp for 3d classification and segmentation of point sets. *arXiv preprint arXiv:2410.09462*, 2024.

Hongyu Sun, Yongcai Wang, Wang Chen, Haoran Deng, and Deying Li. Parameter-efficient prompt learning for 3d point cloud understanding. In *IEEE International Conference on Robotics and Automation*, pages 9478–9486, 2024.

Yue Wang, Yongbin Sun, Ziwei Liu, Sanjay E Sarma, Michael M Bronstein, and Justin M Solomon. Dynamic graph cnn for learning on point clouds. *ACM Transactions on Graphics*, 38(5):1–12, 2019.

Ziyi Wang, Xumin Yu, Yongming Rao, Jie Zhou, and Jiwen Lu. P2p: Tuning pre-trained image models for point cloud analysis with point-to-pixel prompting. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

Wenxuan Wu, Zhongang Qi, and Li Fuxin. Pointconv: Deep convolutional networks on 3d point clouds. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

Xiaoyang Wu, Yixing Lao, Li Jiang, Xihui Liu, and Hengshuang Zhao. Point transformer v2: Grouped vector attention and partition-based pooling. In *Advances in Neural Information Processing Systems*, volume 35, pages 33330–33342, 2022.

Xiaoyang Wu, Li Jiang, Peng-Shuai Wang, Zhijian Liu, Xihui Liu, Yu Qiao, Wanli Ouyang, Tong He, and Hengshuang Zhao. Point transformer v3: Simpler faster stronger. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4840–4851, 2024.

Tiange Xiang, Chaoyi Zhang, Yang Song, Jianhui Yu, and Weidong Cai. Walk in the cloud: Learning curves for point clouds shape analysis. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

Mutian Xu, Junhao Zhang, Zhipeng Zhou, Mingye Xu, Xiaojuan Qi, and Yu Qiao. Learning geometry-disentangled representation for complementary understanding of 3d object point cloud. In *AAAI Conference on Artificial Intelligence*, 2021.

Xumin Yu, Lulu Tang, Yongming Rao, Tiejun Huang, Jie Zhou, and Jiwen Lu. Point-bert: Pre-training 3d point cloud transformers with masked point modeling. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19313–19322, 2022.

Yaohua Zha, Jinpeng Wang, Tao Dai, Bin Chen, Zhi Wang, and Shu-Tao Xia. Instance-aware dynamic prompt tuning for pre-trained point cloud models. In *IEEE/CVF International Conference on Computer Vision*, pages 14161–14170, 2023.

Zhengli Zhai, Xin Zhang, and Luyao Yao. Multi-scale dynamic graph convolution network for point clouds classification. *IEEE Access*, 9:62041–62050, 2021.

Cheng Zhang, Haocheng Wan, Xinyi Shen, and Zizhao Wu. Pvt: Point-voxel transformer for point cloud learning. *arXiv preprint arXiv:2108.06076*, 2021.

Renrui Zhang, Ziyu Guo, Peng Gao, Rongyao Fang, Bin Zhao, Dong Wang, Yu Qiao, and Hongsheng Li. Point-m2ae: multi-scale masked autoencoders for hierarchical point cloud pre-training. In *Advances in Neural Information Processing Systems*, volume 35, pages 27061–27074, 2022.

Xiangdong Zhang, Shaofeng Zhang, and Junchi Yan. Pcp-mae: Learning to predict centers for point masked autoencoders. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Hengshuang Zhao, Li Jiang, Chi-Wing Fu, and Jiaya Jia. PointWeb: Enhancing Local Neighborhood Features for Point Cloud Processing. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5560–5568, 2019.

Hengshuang Zhao, Li Jiang, Jiaya Jia, Philip HS Torr, and Vladlen Koltun. Point transformer. In *IEEE/CVF International Conference on Computer Vision*, pages 16259–16268, 2021.

Xin Zhou, Dingkang Liang, Wei Xu, Xingkui Zhu, Yihan Xu, Zhikang Zou, and Xiang Bai. Dynamic adapter meets prompt tuning: Parameter-efficient transfer learning for point cloud analysis. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14707–14717, 2024.

Appendix A. Supported Models

The complete 56-configuration support matrix is provided in the online repository.

Appendix B. SSL Audit Summary

Each of the seven SSL methods was audited in two independent rounds against its original GitHub repository. The audit verified (i) the core encoder/decoder architecture and loss functions. (ii)checkpoint loading with correct prefix-stripping; (iii) Parameter freezing behavior during fine-tuning (iv) Classification and segmentation output consistency. All 7 SSL methods were verified as being faithful to their original implementations after the fixes were applied.

Appendix C. Application: LiDAR Tree Species Classification

As a case study, we evaluated all 56 model configurations on the STPCTLS terrestrial laser scanning dataset Seidel et al. (2021), which contains high-resolution point clouds from artificial and natural forests in Germany and the United States. Seven tree species—beech, red oak, ash, oak, douglas fir, spruce, and pine—were selected for their significant morphological similarity across species and considerable within-species variation owing to diverse growth conditions, making the classification task challenging. To further increase the difficulty, we added isotropic Gaussian noise ($\sigma = 0.20$) to the normalized point clouds. The dataset was divided into training and testing sets using an 80:20 ratio. The sample sizes and class distributions are presented in Table 2.

Table 2: Train and test distribution for the STPCTLS dataset.

Species	**Total**	**Train**	**Test**	**%**
Douglas Fir (*Pseudotsuga menziesii*)	183	146	37	26.5
Beech (*Fagus sylvatica*)	164	131	33	23.7
Spruce (*Picea abies*)	158	126	32	22.9
Red Oak (*Quercus rubra*)	100	80	20	14.5
Ash (*Fraxinus excelsior*)	39	31	8	5.6
Pine (*Pinus sylvestris*)	25	20	5	3.6
Oak (*Quercus robur*)	23	18	4	3.2
Total	**692**	**552**	**139**	**100.0**

Tables 3 and 4 are auto-generated from the cross-validation summaries via `scripts/generate_comparison_table_one_table.py`. Table 3 presents classification accuracy, balanced accuracy, macro F1, recall, and precision for all 56 model configurations, grouped by category. Among the supervised methods, PointSCNet Chen et al. (2022) and PointNet++ (MSG) Qi et al. (2017b) achieve the best performance. Among the SSL backbones, ACT with PointGST fine-tuning achieved the highest overall accuracy (69.78The per-class recall (Table 4) reveals that minority species (Ash, Oak, and Pine) remain challenging for all methods, with the best per-class recall on Ash reaching only 50

Table 3: STPCTLS classification results: performance metrics and model parameters.

Category	Model	Strategy	Accuracy (%)	Balanced Acc (%)	F1 Macro (%)	Recall (%)	Precision (%)	Total Params (M)	Train Params (M)	Epoch Time (s)
Point-based	PointNet	-	51.08	46.02	43.61	46.02	47.36	3.47	3.47	0.96
	PointNet2-SSG	-	56.12	46.62	46.70	46.62	58.99	1.46	1.46	2.78
	PointNet2-MSG	-	67.63	**65.24**	65.60	**65.24**	68.41	1.73	1.73	8.50
	SONet	-	41.01	24.26	22.41	24.26	24.79	2.66	2.66	0.99
	PPFNet	-	64.03	63.96	65.35	63.96	69.75	0.28	0.28	2.37
	PointCNN	-	54.68	47.23	50.42	47.23	59.79	0.27	0.27	9.84
	PointWeb	-	32.37	19.16	16.13	19.16	14.59	0.78	0.78	26.12
	PointConv	-	50.36	47.45	43.85	47.45	44.86	19.56	19.56	7.05
	RSCNN	-	48.20	30.08	25.90	30.08	30.43	1.28	1.28	1.79
	PointMLP	-	48.92	40.61	41.27	40.61	48.72	13.23	13.23	13.07
	PointSCNet	-	69.06	64.27	**67.70**	64.27	**73.53**	1.82	1.82	5.77
	RepSurf	-	67.63	59.61	59.47	59.61	59.74	1.47	1.47	4.92
	PointKAN	-	48.92	41.64	43.27	41.64	56.17	**0.16**	0.16	**0.75**
	DELA	-	32.37	17.98	11.91	17.98	9.60	5.33	5.33	1.41
Attention-based	PCT	-	46.04	36.64	36.71	36.64	37.76	2.87	2.87	3.49
	P2P	-	31.65	17.79	13.86	17.79	18.11	25.04	25.04	11.08
	PointTNT	-	57.55	48.74	45.45	48.74	44.70	3.93	3.93	3.15
	GlobalTransformer	-	58.27	53.45	51.84	53.45	55.14	3.75	3.75	4.01
	PVT	-	61.15	53.08	50.81	53.08	60.61	9.16	9.16	18.97
	PointTransformer	-	59.71	56.62	52.77	56.62	51.27	3.26	3.26	2.07
	PointTransformerV2	-	59.71	54.75	53.54	54.75	64.53	9.59	9.59	9.41
	PointTransformerV3	-	50.36	41.66	37.40	41.66	36.80	12.88	12.88	7.70
Graph-based	DGCNN	-	58.99	59.35	54.11	59.35	53.99	1.80	1.80	4.63
	DeepGCN	-	57.55	53.27	51.40	53.27	50.98	2.21	2.21	14.91
	CurveNet	-	61.87	58.57	58.71	58.57	59.66	2.12	2.12	5.96
	GDAN	-	60.43	53.55	51.90	53.55	52.52	0.93	0.93	7.40
	MS-DGCNN	-	58.99	56.15	54.33	56.15	55.89	1.55	1.55	5.44
	KAN-DGCNN	-	56.12	48.83	44.92	48.83	48.63	1.40	1.40	4.10
	MS-DGCNN++	-	57.55	52.61	53.28	52.61	61.64	1.81	1.81	6.91
Self-supervised	PointMAE	FF	63.31	60.88	57.47	60.88	57.92	22.09	22.09	2.16
		DAPT	53.24	49.70	49.30	49.70	52.89	22.89	1.06	2.04
		IDPT	64.75	61.39	59.97	61.39	60.06	23.52	1.70	1.82
		PPT	56.83	48.21	47.78	48.21	48.56	22.78	1.04	2.77
		GST	64.75	60.18	58.99	60.18	58.71	22.45	0.62	2.82
		VPT	53.96	51.60	53.67	51.60	59.38	22.18	0.36	1.31
	ACT	FF	46.76	30.65	28.23	30.65	27.87	22.09	22.09	2.17
		DAPT	53.96	48.36	44.47	48.36	43.72	22.89	1.06	2.06
		IDPT	57.55	47.34	47.14	47.34	58.77	23.52	1.70	1.97
		PPT	61.87	54.35	53.17	54.35	57.20	22.78	1.04	3.00
		GST	**69.78**	64.09	64.21	64.09	65.21	22.45	0.62	2.88
		VPT	56.12	49.77	49.02	49.77	60.29	22.18	0.36	1.31
	RECON	FF	39.57	26.56	26.67	26.56	38.01	43.12	**0.01**	2.18
		DAPT	53.24	49.09	48.85	49.09	51.29	22.89	1.06	1.97
		IDPT	58.27	48.88	47.24	48.88	46.09	23.52	1.70	1.98
		PPT	57.55	58.02	56.06	58.02	56.17	22.78	1.04	2.82
		GST	67.63	60.15	63.19	60.15	69.53	22.45	0.62	2.63
		VPT	54.68	50.93	48.86	50.93	53.25	22.18	0.36	1.33
	PointGPT	FF	61.87	52.45	52.99	52.45	54.99	29.23	29.23	2.66
		DAPT	61.15	53.34	53.21	53.34	53.42	22.78	0.95	1.94
		IDPT	61.87	48.40	50.13	48.40	58.41	23.49	1.70	2.10
		PPT	61.87	54.60	55.80	54.60	61.33	22.78	0.99	3.48
		GST	62.59	52.87	52.41	52.87	54.96	22.45	0.62	2.87
		VPT	50.36	37.53	35.64	37.53	36.05	22.15	0.36	1.44
	PointM2AE	FF	61.87	57.05	53.92	57.05	52.96	12.82	12.82	5.98
	PointBERT	FF	61.87	56.02	56.37	56.02	58.71	22.06	22.06	2.11
	PCP	FF	46.04	38.83	33.23	38.83	35.67	22.34	22.34	2.40

Table 4: STPCTLS per-class recall (%).

Category	Model	Strategy	Rec(Buche) (%) ↓ n=33	Rec(Douglasie) (%) ↓ n=37	Rec(Eiche) (%) ↓ n=4	Rec(Esche) (%) ↓ n=8	Rec(Fichte) (%) ↓ n=32	Rec(Kiefer) (%) ↓ n=5	Rec(Roteiche) (%) ↓ n=20
Point-based	PointNet	-	54.55	43.24	75.00	12.50	71.88	20.00	45.00
	PointNet2-SSG	-	78.79	59.46	50.00	37.50	65.62	20.00	15.00
	PointNet2-MSG	-	75.76	45.95	75.00	50.00	75.00	40.00	**95.00**
	SONet	-	60.61	64.86	0.00	0.00	34.38	0.00	10.00
	PPFNet	-	81.82	70.27	**100.00**	50.00	40.62	40.00	65.00
	PointCNN	-	69.70	54.05	50.00	12.50	59.38	40.00	45.00
	PointWeb	-	12.12	21.62	0.00	0.00	12.50	20.00	0.00
	PointWeb	-	45.45	32.43	0.00	0.00	56.25	0.00	0.00
	PointConv	-	54.55	43.24	75.00	12.50	71.88	40.00	35.00
	RSCNN	-	93.94	21.62	0.00	0.00	75.00	0.00	20.00
	PointMLP	-	60.61	40.54	25.00	0.00	53.12	40.00	65.00
	PointSCNet	-	**93.94**	54.05	75.00	50.00	71.88	40.00	65.00
	RepSurf	-	75.76	62.16	50.00	37.50	71.88	40.00	80.00
	PointKAN	-	42.42	54.05	50.00	12.50	62.50	20.00	50.00
	DELA	-	39.39	**86.49**	0.00	0.00	0.00	0.00	0.00
Attention-based	PCT	-	63.64	45.95	0.00	25.00	46.88	40.00	35.00
	P2P	-	36.36	75.68	0.00	0.00	12.50	0.00	0.00
	PointTNT	-	57.58	64.86	75.00	0.00	68.75	20.00	55.00
	GlobalTransformer	-	54.55	78.38	75.00	12.50	43.75	40.00	70.00
	PVT	-	78.79	62.16	75.00	12.50	53.12	20.00	70.00
	PointTransformer	-	72.73	56.76	75.00	12.50	59.38	60.00	60.00
	PointTransformerV2	-	72.73	56.76	75.00	12.50	56.25	40.00	70.00
	PointTransformerV3	-	90.91	43.24	75.00	0.00	37.50	0.00	45.00
Graph-based	DGCNN	-	48.48	51.35	75.00	12.50	78.12	**80.00**	70.00
	DeepGCN	-	57.58	54.05	75.00	12.50	68.75	40.00	65.00
	CurveNet	-	69.70	54.05	75.00	37.50	68.75	40.00	65.00
	GDAN	-	72.73	70.27	75.00	0.00	46.88	40.00	70.00
	MS-DGCNN	-	51.52	62.16	75.00	12.50	71.88	60.00	60.00
	KANDGCNN	-	42.42	83.78	75.00	0.00	40.62	20.00	80.00
	MS-DGCNN++	-	69.70	81.08	75.00	12.50	25.00	40.00	65.00
Self-supervised	PointMAE	FF	57.58	64.86	75.00	25.00	68.75	60.00	75.00
		DAPT	57.58	45.95	75.00	12.50	71.88	40.00	45.00
		IDPT	57.58	70.27	75.00	25.00	71.88	60.00	70.00
		PPT	63.64	59.46	50.00	25.00	59.38	20.00	60.00
		GST	69.70	54.05	50.00	37.50	75.00	60.00	75.00
		VPT	51.52	45.95	50.00	25.00	68.75	60.00	60.00
	ACT	FF	45.45	37.84	0.00	0.00	81.25	0.00	50.00
		DAPT	51.52	43.24	75.00	0.00	68.75	20.00	80.00
		IDPT	57.58	59.46	50.00	12.50	71.88	20.00	60.00
		PPT	54.55	70.27	50.00	12.50	78.12	60.00	55.00
		GST	66.67	67.57	50.00	**50.00**	84.38	60.00	70.00
		VPT	54.55	51.35	75.00	12.50	75.00	20.00	60.00
	RECON	FF	45.45	64.86	25.00	0.00	40.62	0.00	10.00
		DAPT	54.55	37.84	75.00	0.00	81.25	40.00	55.00
		IDPT	66.67	56.76	50.00	0.00	68.75	40.00	60.00
		PPT	54.55	45.95	75.00	12.50	78.12	80.00	60.00
		GST	81.82	72.97	50.00	50.00	56.25	40.00	70.00
		VPT	54.55	43.24	75.00	12.50	81.25	40.00	50.00
	PointGPT	FF	81.82	45.95	25.00	37.50	71.88	40.00	65.00
		DAPT	81.82	54.05	50.00	25.00	62.50	40.00	60.00
		IDPT	66.67	62.16	25.00	25.00	75.00	20.00	65.00
		PPT	78.79	54.05	50.00	25.00	59.38	40.00	75.00
		GST	72.73	72.97	25.00	37.50	46.88	40.00	75.00
		VPT	78.79	27.03	0.00	37.50	59.38	0.00	60.00
	PointM2AE	FF	57.58	56.76	50.00	12.50	**87.50**	80.00	55.00
	PointBERT	FF	66.67	64.86	50.00	25.00	65.62	60.00	60.00
	PCP	FF	45.45	35.14	0.00	0.00	81.25	80.00	30.00